

PAPER_372 — Compressed UQFF with B/Bcrit Superconductivity

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Star Magic UQFF Whitepaper Series

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Abstract

This paper presents the Compressed UQFF formulation, a multi-term master gravity equation that incorporates Newtonian gravity, Hubble expansion, superconductivity via the B/Bcrit flux-quenching factor, environmental coupling, cosmological constant contribution, quantum coherence, fluid dynamics, and dark matter perturbation. The framework is parameterised for seven astrophysical systems and validated via unit test against SGR1745-2900. This is the first UQFF formulation to explicitly incorporate Bardeen-Cooper-Schrieffer (BCS) superconductivity quenching into the gravitational coupling (Linear Meissner form; see PAPER_375 for the exponential form).

1. Master Equation

$$\begin{aligned} g_{\mathrm{UQFF}}(r,t) = & \frac{G M(t)}{r^2} \cdot [1 + H(t,z)] \cdot \left[1 - \frac{B}{B_{\mathrm{crit}}}\right] \cdot [1 + F_{\mathrm{env}}] \\ & + \frac{(\rho_{g1} + \rho_{g2} + \rho_{g3}' + \rho_{g4})}{\Lambda c^2} \\ & + \frac{\hbar \Delta x \cdot \Delta p}{\int \psi_{\mathrm{total}}^* \hat{H} \psi_{\mathrm{total}} dV} \cdot \frac{2\pi}{t_{\mathrm{Hubble}}} \\ & + \rho_{\mathrm{fluid}} V g \\ & + (M_{\mathrm{vis}} + M_{\mathrm{DM}}) \left(\frac{\Delta \rho}{\rho} + \frac{3GM}{r^3} \right) \end{aligned}$$

where $H(t,z) = H_0 t$ (Newtonian cosmological expansion approximation), $H_0 = 2.269 \times 10^{-18} \text{ s}^{-1}$.

2. Modular Component Functions

Function	Formula	Constants
compressed_base	GM/r^2	$G = 6.674 \times 10^{-11}$
compressed_expansion	$1 + H_0 t$	$H_0 = 2.269 \times 10^{-18} \text{ s}^{-1}$
compressed_super_adj	$1 - B/B_{\mathrm{crit}}$	linear Meissner
compressed_env	1.0	default

Function	Formula	Constants
compressed_cosm	$\Lambda c^2/3$	$\Lambda = 1.1 \times 10^{-2} \text{ m}^{-2}$
compressed_quantum	$(\hbar/10^{-68}) \cdot 2.176 \times 10^{-18} \cdot (2\pi/t_H)$	$t_H = 4.35 \times 10^1 \text{ s}$
compressed_fluid	$\rho_f v_{gl}$	from MUGESystem
compressed_perturbation	$(M+M_{DM})(\delta\rho/\rho + 3GM/r^3)$	$\delta\rho/\rho = 10^{-}$

3. System Parameters

System	M (kg)	r (m)	B (T)	Bcrit (T)	Vsys (m³)	ffluid (Hz)
Magnetar SGR1745-2900	2.984×10^{30}	$1 \times 10^$	1×10^{10}	1×10^{11}	4.189×10^{12}	1.269×10^{-1}
Sagittarius A*	8.155×10^3	1×10^{12}	$1 \times 10^{-}$	$1 \times 10^{-}$	$3.552 \times 10^{}$	$3.465 \times 10^{-}$
Tapestry Starbirth	1.989×10^3	3.086×10^1	$1 \times 10^{-}$	1×10^{-3}	1×10^3	1×10^{-12}
Westerlund 2	1.989×10^3	3.086×10^1	$1 \times 10^{-}$	1×10^{-3}	1×10^3	1×10^{-12}
Pillars of Creation	1.989×10^{32}	9.46×10^1	$1 \times 10^{-}$	1×10^{-3}	$3.552 \times 10^{}$	8.457×10^{-1}
Rings of Relativity	1.989×10^3	3.086×10^1	$1 \times 10^{-}$	$1 \times 10^{-}$	$1 \times 10^{}$	$1 \times 10^{-}$
Student's Guide Universe	1×10^3	1×10^2	1×10^{-10}	$1 \times 10^{-}$	1×10^0	1×10^{-1}

4. Validation

Unit test: test_compute_compressed_MUGE(SGR1745-2900)

Expected: $1.782 \times 10^3 \text{ m/s}^2$

(Dominated by compressed_base × expansion; B/Bcrit = 0.1 → 90% retention)

5. Physical Interpretation

The $[1 - B/B_{\mathrm{crit}}]$ factor models the Meissner effect: as the magnetic field B approaches the critical field Bcrit, gravitational coupling is quenched—mirroring how a Type-I superconductor expels magnetic flux below Bcrit. The compressed UQFF thus unifies gravitomagnetic quenching with cosmological expansion and quantum corrections in a single framework. (For Type-II exponential treatment, see PAPER_375.)

6. Implementation

C++: STAR_MAGIC_09SEPT_UQFF_MODULE.cpp, namespace CompressedUQFF **Python:** CondensedPhysics4.py, class CompressedUQFFBcritSuperconductivityCalculator (CP4 #20) **WOLFRAM_TERM:** WOLFRAM_TERM_COMPRESSED_UQFF_BCRIT

